

Session 5

JSP Expression Language



Review 2-1

- ❑ JavaBeans are reusable components that can be used in multiple applications.
- ❑ Bean properties can be accessed using the `get()` and `set()` methods.
- ❑ JSP pages can include JavaBean components using Standard Actions.
- ❑ JSP Standard Actions are used to insert files and bean components, and forward the user to another Web page.

Review 2-2

- ☐ `useBean` action makes a JavaBean component available in a JSP page.
- ☐ `setProperty` action sets a value to the property of the bean.
- ☐ `getProperty` action is used to get a value from the bean.
- ☐ `Forward` action will forward the request of the user to another page.
- ☐ `Include` action is used to include a page in the current JSP page.

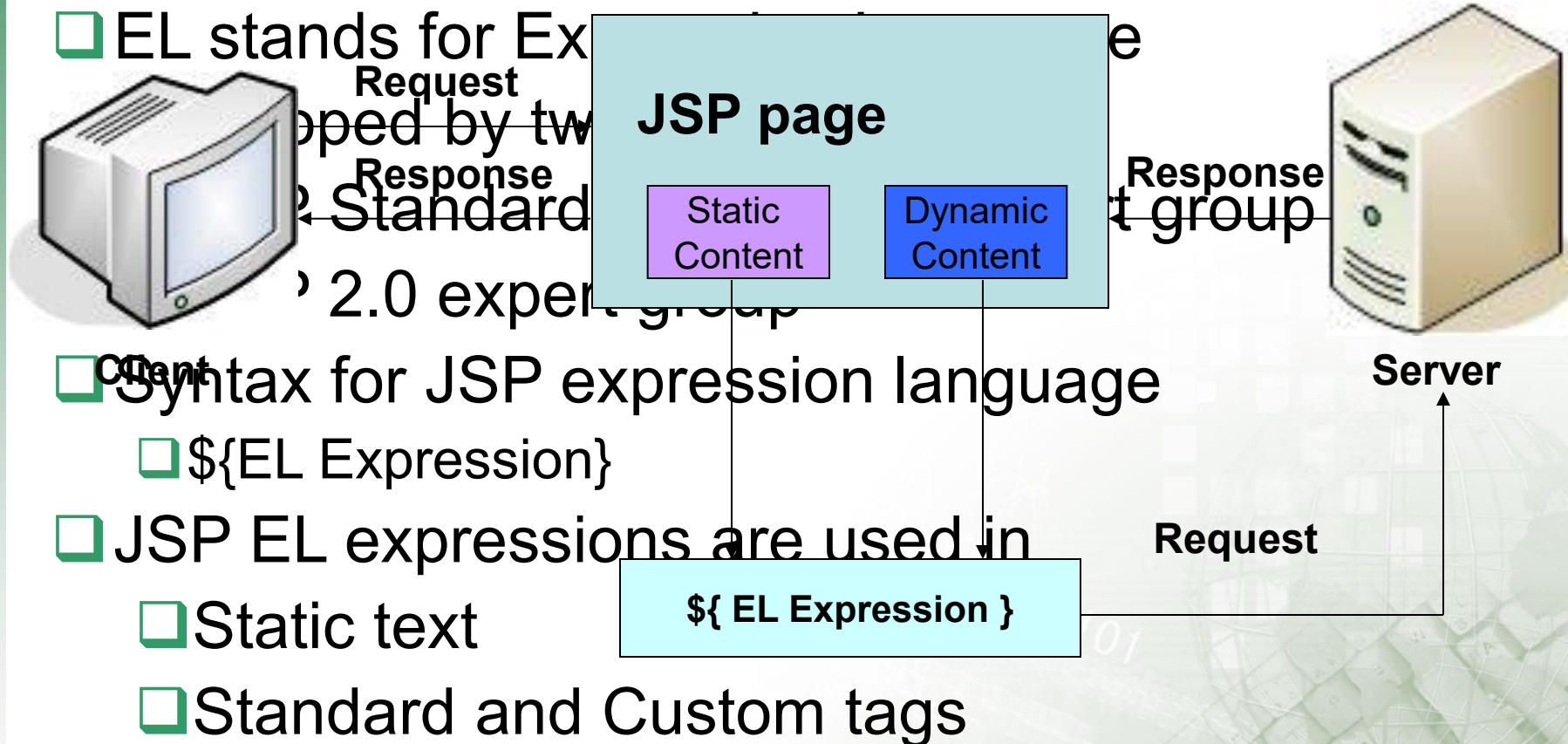
Objectives

- ❑ Describe the syntax of JSP Expression Language.
- ❑ Identify the advantages of using JSP
- ❑ Use Expression Language in JSP

Expression Language

- ❑ EL simplifies the accessibility of data stored in the Java Bean Component .
- ❑ And other objects like request, session, application etc..
- ❑ There are many implicit objects, operators & reserved words in EL.

Introduction to EL 4-1



Introduction to EL 4-2

```
<html>
  <head>
    <title>Expression Language</title>
  </head>
  <body>
    <h1>EL operations</h1>
    <table border="1">
      <tr>
        <td><b>Concept</b></td>
        <td><b>EL Expression</b></td>
        <td><b>Result</b></td>
      </tr>
      <tr>
        <td>And</td>
        <td>${'${'}true and true}</td>
        <td>${true and true}</td>
      </tr>
      <tr>
        <td>And</td>
        <td>${'${'}true && false}</td>
        <td>${true && false}</td>
      </tr>
    </table>
  </body>
</html>
```

EL expressions in
<table> tags

EL expressions in
<table> tags

D

Introduction to EL 4-3

```
<html>
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-
8">
    <title>Article Expression</title>
  </head>
  <body>
    <h1>Article Expression</h1>
    <h2>Price of the article is ${84.5E4}.</h2>
  </body>
</html>
```



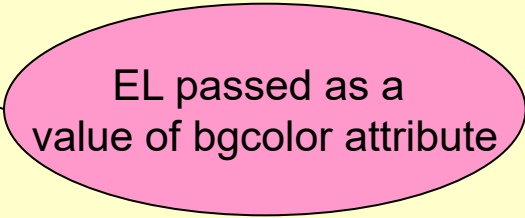
EL expressions in
<h2> tags (Static text)

The diagram shows a pink oval containing the text "EL expressions in <h2> tags (Static text)". An arrow points from this oval to a red dashed box that highlights the EL expression "\${84.5E4}" within the <h2> tag in the HTML code above.

Demonstration: Example 2

Introduction to EL 4-4

```
<html>
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    <title>EL Example</title>
  </head>
  <%
    pageContext.setAttribute("color", "#FFFFCC");
  %>
  <body bgcolor='${pageScope.color}'>
    <h1>
      Background color changed
    </h1>
  </body>
</html>
```



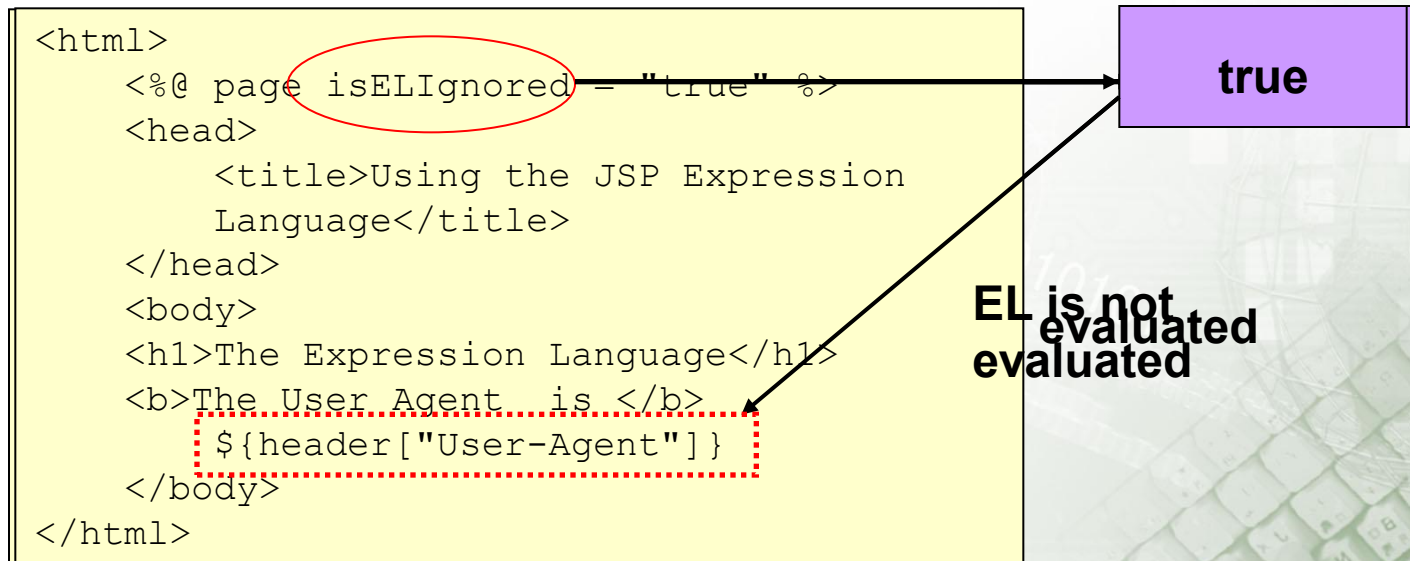
EL passed as a value of bgcolor attribute

Demonstration: Example 3

Disabling the EL 2-1

- ❑ Disable evaluation to ignore the execution of EL expression in JSP
- ❑ Syntax to disable EL evaluation

`<%@ page isELIgnored ="true|false" %>`



Disabling the EL 2-2

```
<%@page isELIgnored="true" %>
```

isELIgnored
set to true

```
<html>
```

```
<head>
```

```
<title>ELDisabled</title>
```

```
</head>
```

```
<body bgcolor="#ffffff">
```

```
<h1>
```

```
Enabling/Disabling Expression Language
```

```
</h1>
```

```
<form method="post" action="ELDisabled.jsp">
```

```
<b> ${'First Name: '}<input type="text"  
value="${'Enter your first name'}"/> </b><br />
```

```
<b> ${'Last Name: '}<input type="text"  
value="${'Enter your last name'}"/> </b><br />
```

```
<hr/>
```

```
<input type="submit" name="Submit" value="${'Submit'}">
```

```
<input type="reset" value="${'Reset'}">
```

```
</form>
```

```
</body>
```

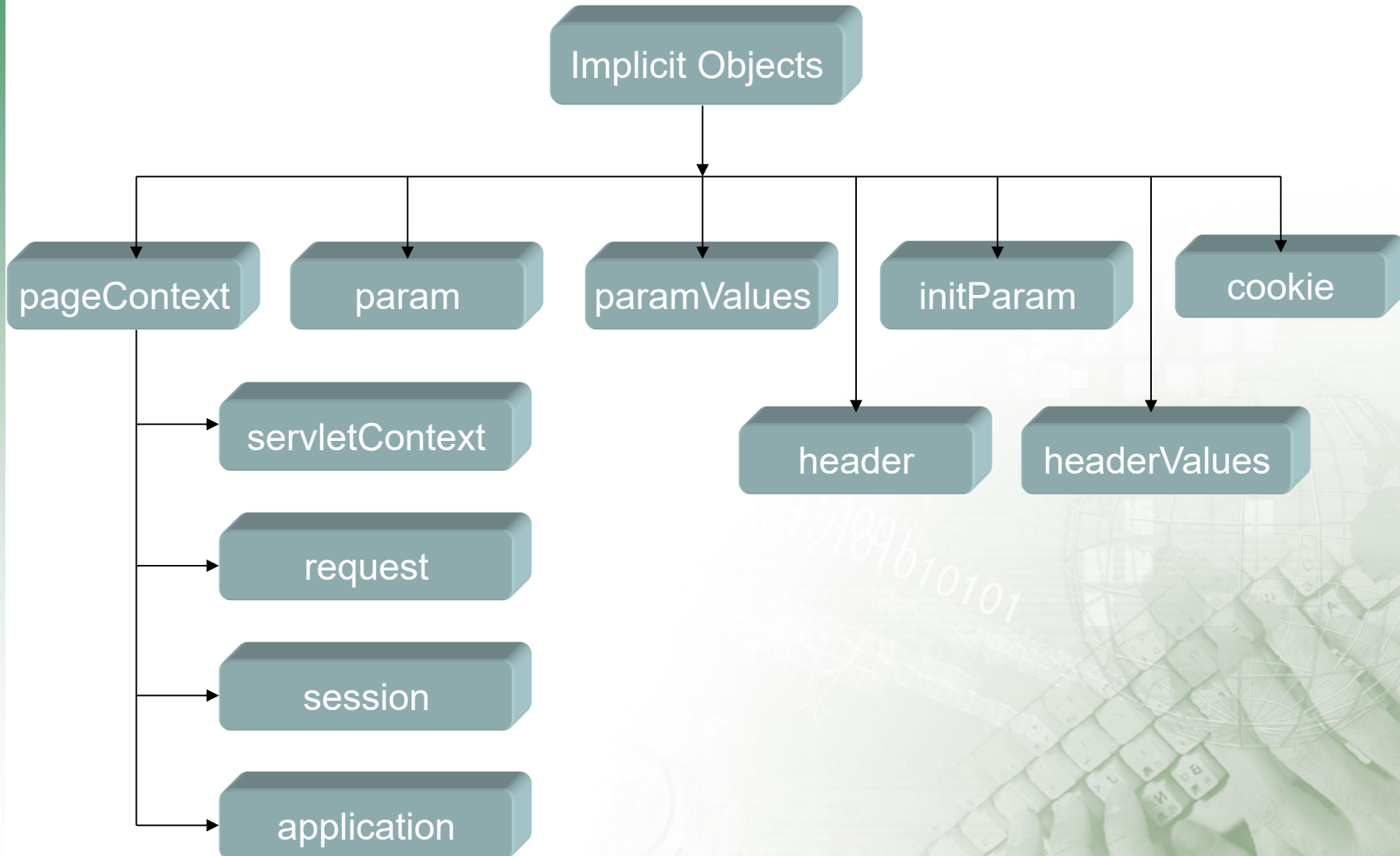
```
</html>
```

JSP Engine ignores
EL expressions

Variables

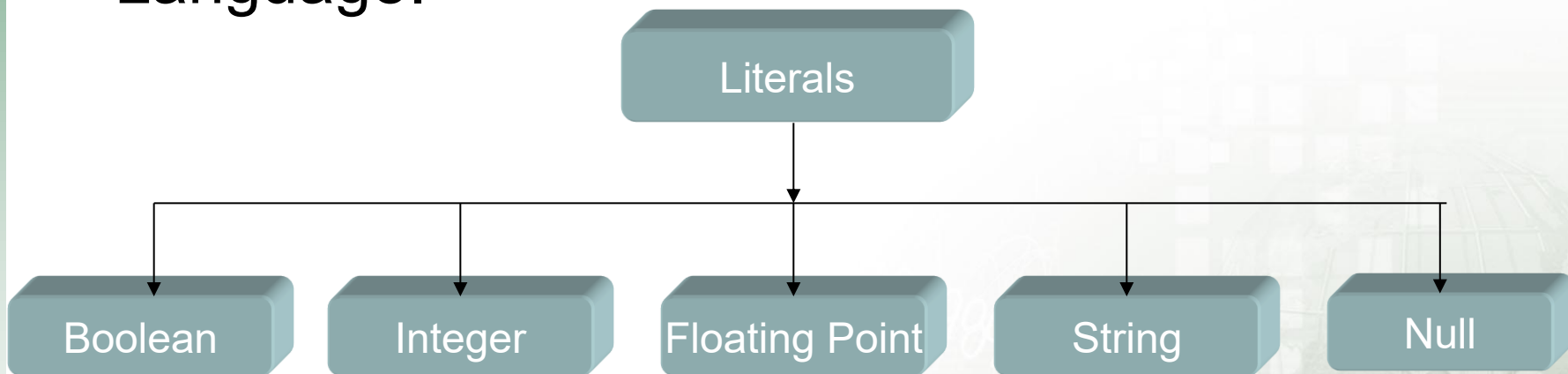
- ❑ Web container evaluates a variable
- ❑ Value of variable is determined using `pageContext.findAttribute(String)`
- ❑ Dot operator (`.`) or square brackets(`[ ]`) can be used to access value of variable

Implicit Objects

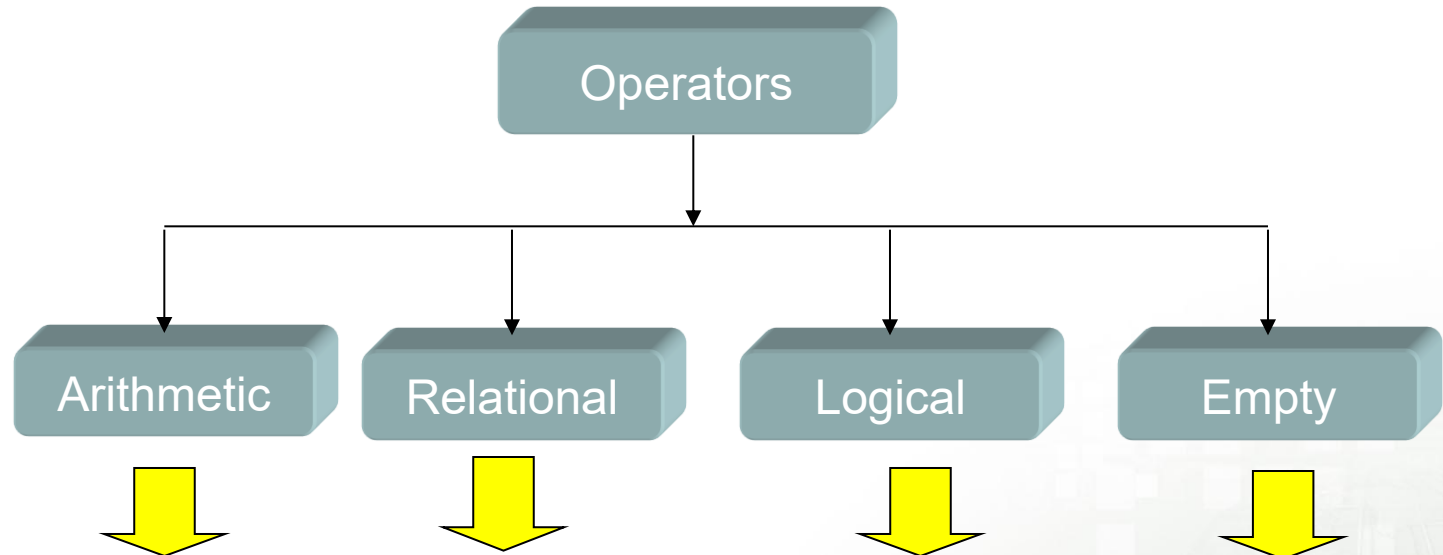


Literals

- ❑ Literals are assigned fixed values
- ❑ Literals are supported by JSP Expression Language.



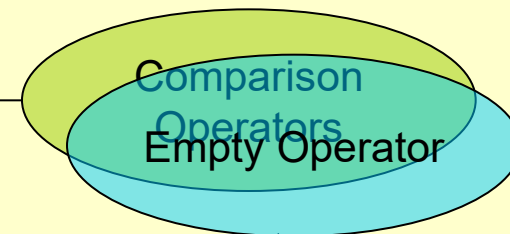
Operators 3-1



Operator	Operators		Operators	Operators	Description
*	< or lt	Multiplication	Less than	&& or and	Logical AND
/ or div	> or gt	Division	Greater than	or or	Logical OR
+	< = or le	Addition	Less than or equal to		
-	> = or ge	Subtraction	Greater than or equal to		
	= =	Equality			
	!=	Inequality			

Operators 3-2

```
<html>
  <head>
    <meta http-equiv="Content-Type" content="text/html;
charset=UTF-8">
    <title>Operators</title>
  </head>
  <body>
    <h1>Operators</h1>
    <b>
      4 &gt; 3 ${4 > '3'}<br/>
      4 &lt; 3 ${4 < '3'}<br/>
      4 &ge; 3 ${4 >= 3}<br/>
      4 &le; 3 ${4 le 3} <br/>
      4 = 4 ${4 == 4}
    </b>
    <h1>Empty Operator</h1>
    <b>
      empty "" ${empty ""}<br/>
      empty "sometext" ${empty "sometext"}<br/>
    </b>
  </body>
</html>
```



Operators 3-3

```
<html>
  <head>
    <meta http-equiv="Content-Type" content="text/html;
<body bgcolor='${pageScope.color}' text='${pageScope.textclr}'>
  <h1>Use of implicit objects in EL.</h1>
  <font color="${pageScope.foregr}"
    size="${pageScope.size}">
    Text color and font size changed.
  </font><br/>
  value of num1=${pageScope.num1}<br />
  value of num2=${pageScope.num2}<br />
  value of num1 = <b>${pageScope.num1}</b> * num2 = <b>
    ${pageScope.num2}</b> =<b> ${pageScope.num1 *
    pageScope.num2}</b>
</body>
</html>

<%
  pageContext.setAttribute("foregr", foregr);
%>
```

EL used to access
variable and set attributes

Use of * operator

Demonstration: Example 6

Reserved words

- ❑ Reserved words should not be used in the program as the name of variables or methods.

Operators	Description	Operators	Description
and	And	ge	Greater than or equal to
or	Or	true	True
not	Not	false	False
eq	Equal to	null	Null
ne	Not equal to	empty	Empty
le	Less than	div	Division
gt	Greater than	mod	Modulus

Summary

- ❑ EL stands for Expression Language. JSP expression language is used to access data stored in a JavaBean.
- ❑ JSP expression language can be used in any static text, standard tags, and custom tags.
- ❑ Dot operator (.) or ([]) are used to retrieve the value of variable
- ❑ `pageScope`, `requestScope`, `sessionScope`, and `applicationScope` implicit objects are used to access various scope variables.
- ❑ The `pageContext` object provides access to `servletContext`, `session`, `request`, and `response` objects.
- ❑ JSP expression language defines reserved words, such as `and`, `or`, `not`, and `eq`.